

CSA5103-Cryptography and network security

C01-Assessment Tool 1

Scenario based problems

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Scenario 1:

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, the message "**MEET AT NOON**" is transformed into an encoded version using a fixed shift pattern.

- a) Reconstruct the encrypted message if a shift of 4 positions is applied.**
- b) Explain how an authorized member retrieves the original message.**

Ans)

Explanation

To secure the confidential event details, the organization uses the **Caesar Cipher**, one of the oldest and simplest encryption techniques. In this method, every letter in the original message is shifted by a fixed number of positions in the alphabet. Here, the shift value is **+4**, meaning each letter moves four places forward. This process converts the readable message (plaintext) into an unreadable message (ciphertext), ensuring that only authorized members who know the shift value can understand it.

- a) Reconstruct the encrypted message if a shift of 4 positions is applied.**

The Caesar Cipher is a classical encryption technique in which each letter of the plaintext is replaced by another letter that is 4 positions ahead in the English alphabet. This process converts the original readable message into an unreadable ciphertext. Spaces are not changed

Plaintext: MEET AT NOON

Step-by-Step Conversion:

Plain Text	Alphabet Position	Shift (+4)	Cipher Text
M	13	17	Q
E	5	9	I
E	5	9	I
T	20	24	X
A	1	5	E
T	20	24	X
N	14	18	R
O	15	19	S
O	15	19	S
N	14	18	R

Ciphertext:
QIIX EX RSSR

Explanation

- **M** is shifted 4 places forward → **Q**
- **E** is shifted 4 places forward → **I**
- **T** is shifted 4 places forward → **X**
- **A** is shifted 4 places forward → **E**

- **N** is shifted 4 places forward → **R**
- **O** is shifted 4 places forward → **S**

After applying the same **+4 shift** to every letter, the original plaintext "**MEET AT NOON**" is converted into the ciphertext "**QIIX EX RSSR**". This encrypted message can only be understood by someone who knows the shift key and can reverse the process by shifting each letter **4 positions backward**.

1(b) Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Explanation

An authorized member who knows the **Caesar Cipher encryption rule** and the **shift key (+4)** can easily recover the original message through the **decryption process**. Decryption is the reverse of encryption. Instead of moving each letter **4 positions forward**, the receiver moves every letter **4 positions backward** in the alphabet.

Decryption Formula

Plaintext = Ciphertext – Key (4)

or

$P = (C - K) \bmod 26$

Where:

- **P** = Plaintext
- **C** = Ciphertext
- **K** = Shift Key (4)
- **mod 26** ensures the alphabet wraps around correctly.

Step-by-Step Decryption

Cipher Text	Shift (-4)	Plain Text
Q	Q → M	M
I	I → E	E
I	I → E	E
X	X → T	T
E	E → A	A
X	X → T	T
R	R → N	N
S	S → O	O
S	S → O	O
R	R → N	N

Recovered Plaintext

MEET AT NOON

Scenario2:

A cybersecurity intern intercepts the coded message **"GSRH RH Z HVXIVG"** during network monitoring. The structure of the text suggests that it uses a **simple substitution cipher** rather than a complex encryption algorithm.

- a) Deduce the most likely original message.
- b) Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Explanation

To identify the original message, the intern analyzes the pattern of the encrypted text. Since the letters do not appear to follow a fixed shift (like the Caesar Cipher), the message is suspected to use the **Atbash Cipher**, a classical substitution cipher. In the Atbash Cipher, each letter of the alphabet is replaced by its opposite letter. For example, **A is replaced by Z, B by Y, C by X, D by W**, and so on. By applying this reverse alphabet mapping to every letter in the ciphertext, the original plaintext can be recovered. This method helps determine the hidden message and understand the encryption technique used.

a) Deduce the most likely original message.

Step-by-Step Decryption (Atbash Cipher)

Cipher Letter	Atbash Mapping	Plain Letter
G	G → T	T
S	S → H	H
R	R → I	I
H	H → S	S
R	R → I	I
H	H → S	S
Z	Z → A	A
H	H → S	S
V	V → E	E
X	X → C	C
I	I → R	R
V	V → E	E
G	G → T	T

Recovered Plaintext:

THIS IS A SECRET

Explanation:

In the **Atbash Cipher**, each letter is replaced by its opposite letter in the English alphabet (A ↔ Z, B ↔ Y, C ↔ X, D ↔ W, ..., G ↔ T, H ↔ S, I ↔ R). By applying this reverse alphabet mapping to each letter of the ciphertext **"GSRH RH Z HVXIVG"**, the original plaintext **"THIS IS A SECRET"** is obtained. This demonstrates that the message was encrypted using the Atbash substitution rule, allowing the receiver to recover the original message correctly.

b) Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Explanation

The intercepted message **"GSRH RH Z HVXIVG"** follows the characteristics of the **Atbash Cipher**, a classical substitution cipher. Unlike the Caesar Cipher, the Atbash Cipher does not use a shift value. Instead, it replaces every letter with its opposite letter in the English alphabet. For example, **A ↔ Z, B ↔ Y, C ↔ X, D ↔ W**, and so on. By applying this reverse alphabet mapping, the encrypted message is converted back into the original readable message.

The repeated word **"RH"** decrypts to **"IS"**, which is a common English word. Similarly, **"GSRH"** decrypts to **"THIS"**, and **"HVXIVG"** decrypts to **"SECRET"**. These meaningful English words confirm that the message has been encrypted using the Atbash Cipher.

Pattern Analysis:

Cipher Word	Decrypted Word	Observation
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GSRH	THIS	Forms a meaningful English word.
RH	IS	A common English word that appears twice
Z	A	A single-letter English word.
HVXIVG	SECRET	Completes the meaningful sentence.

Cipher Characteristics

Characteristic	Explanation
Cipher Type	Atbash Cipher
Encryption Method	Reverse alphabet substitution (A ↔ Z, B ↔ Y, C ↔ X, ...)
Key Required	No numerical key is used.
Pattern	Each letter is replaced by its opposite alphabet letter.
Final Plaintext	THIS IS A SECRET

Conclusion:

By analyzing the repeating patterns, meaningful word structure, and the reverse alphabet substitution, it is confirmed that the intercepted message uses the **Atbash Cipher**. Therefore, the original message is **"THIS IS A SECRET."** This demonstrates how classical substitution ciphers can be identified and decrypted through pattern analysis and knowledge of known cipher techniques.